Lecture 16: Asset Pricing in High Dimensions

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Intro

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- Last lecture: shrinkage and double selection
- Today: apply those tools to the factor zoo
- Kozak et al. (2020) ask how to estimate an SDF in a huge characteristic space
- Feng et al. (2020) ask whether one new factor still matters after the whole zoo
- Main theme: high dimension changes both estimation and inference

Why This Lecture?

- Lecture 14: weak test assets and multiple testing made the zoo look dangerous (Lewellen et al. 2010; Harvey et al. 2016)
- Lecture 15: regularization and valid post-selection inference gave us new tools (Belloni et al. 2014)
- Today: two modern answers to the same broad problem
- One estimates a robust high-dimensional SDF
- One tests a candidate factor against a high-dimensional benchmark

Flight Plan

- Part I: why Kozak et al. (2020) reject characteristics-sparse factor models
- Part II: how Feng et al. (2020) test whether a new factor adds marginal pricing power
- End: compare the two objects, the two methods, and the two contributions

Part I: Shrinking the Cross-Section

The Question

- Can a few characteristics summarize the whole cross-section?
- FF3 and FF5 made that look plausible (Fama and French 1993, 2015)
- Lecture 14 already made us suspicious (Lewellen et al. 2010; Harvey et al. 2016)
- Kozak et al. (2020) ask: maybe the SDF is not sparse in characteristics at all

Why Sparse Characteristics Are a Strong Claim

- The present-value identity connects observed prices with expected profitability and investment
- Many characteristics can proxy for profitability, investment, distress, ...
- We don't actually observe expectations
- Sparsity in characteristics means extreme redundancy across anomalies

SDF Setup

- Start with a $N \times 1$ vector of excess returns R_t
- Z_t: $N \times H$ matrix of firm characteristics
- $F_t = Z_{t-1}^\top R_t$ are factor portfolios

They consider an SDF of the following form:

$$M_t = 1 - b_{t-1}^\top (R_t - \mathbb{E}_{t-1}[R_t])$$

where $b_{t-1} = Z_{t-1}^\top b$

- If a given $b_j = 0$, then the jth characteristic does not enter the SDF at all
- Sparsity in the SDF is equivalent to sparsity of b

SDF Setup

Plugging one equation into the other, we can rewrite the SDF as

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- A conditional model for the SDF entails $\mathbb{E}_{t-1}[M_t R_t] = 0$
- Then it should be the case that

  $$\mathbb{E}[M_t F_t] = 0$$

  since factors are themselves portfolios of returns!

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- Solving for b gives:

  $$b = \Sigma^{-1}\mu$$

  where $\Sigma = \mathbb{E}[(F_t - \mathbb{E}[F_t])(F_t - \mathbb{E}[F_t])^\top]$ and $\mu = \mathbb{E}[F_t]$

Naive Estimation Fails

$$\hat{b}^{naive} = \hat{\Sigma}_1^{-1}\bar{\mu}$$

- This is just the sample analog of the exact formula
- Problem 1: expected returns are noisy
- Problem 2: H can be large relative to T, screwing up the covariance estimate
- Result: the estimator overfits means and loads too much on weak directions

Rotate to Principal Components

$$\Sigma_1 = QDQ^\top, \qquad P_t = Q^\top F_t, \qquad b_P = D^{-1}\mathbb{E}[P_t]$$

- PCs rank directions by variance
- Low-eigenvalue PCs are statistically fragile
- Directions of low variance generate huge coefficients unless the premia are estimated to be zero
- If we mis-handle them, the estimated SDF becomes wild

The Economic Prior

- Suppose you know Σ
- They assume a Gaussian prior for μ:

$$\mu \sim \mathcal{N}\left(0, \kappa^2 \frac{\Sigma^\eta}{\tau}\right), \qquad \tau = \text{tr}[\Sigma] = \sum_{j=1}^{H} d_j$$

  where d_j is the j-th eigenvalue of Σ
- Recall that $\Sigma^\eta = QD^\eta Q^\top$
- If η is large, directions of low eigenvalues are taken to zero a priori

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- Recall that $\Sigma^\eta = QD^\eta Q^\top$
- If η is large, directions of low eigenvalues are taken to zero a priori
- Notice that $b \sim \mathcal{N}(0, \kappa^2 \frac{\Sigma^{\eta-2}}{\tau})$
- The variance about b is huge if η is too small
- They pick $\eta = 2$

The Shrinkage Estimator

- We get a spherical prior on coefficients: $b \sim \mathcal{N}\left(0, \frac{\kappa^2}{\tau} I_H\right)$
- Now they assume factors come from an i.i.d. Gaussian distribution
- This lets you derive the posterior distribution of b given the *observed* mean of factors:

$$\hat{b} = \left(\Sigma + \gamma I_H\right)^{-1} \bar{\mu}$$

where $\gamma = \frac{\tau}{T \cdot \kappa^2}$

- This looks like something you saw last class, right?

Penalized Interpretation

$$\hat{b} = \arg\min_{b} \left\{ (\bar{\mu} - \Sigma b)^{\top} \Sigma^{-1} (\bar{\mu} - \Sigma b) + \gamma \, b^{\top} b \right\}$$

- That estimator is equivalent to the argmin of the above objective function
- Recall that $\|b\|_2^2 = b^{\top} b$
- This is like Ridge but with a non-standard loss function

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- Recall that $\|b\|_2^2 = b^\top b$
- This is like Ridge but with a non-standard loss function
- Another equivalent optimization problem is

$$\widehat{b} = \arg\min_b \{(\bar{\mu} - \Sigma b)^\top (\bar{\mu} - \Sigma b) + \gamma\, b^\top \Sigma b\}$$

- This is like Ridge but with a different penalty
- $b^\top \Sigma b$ is the maximal Sharpe ratio attained. Why?

Why Add an ℓ_1 Penalty?

$$\hat{b} = \arg\min_b \left\{ (\bar{\mu} - \Sigma_1 b)^\top \Sigma^{-1} (\bar{\mu} - \Sigma_1 b) + \gamma_2\, b^\top b + \gamma_1 \sum_{j=1}^{H} |b_j| \right\}$$

- ℓ_1 can drop useless factors
- But pure lasso is bad with correlated predictors
- And characteristic portfolios are very correlated
- This is like the Elastic Net, with a non-standard loss function
- Their main empirical message: ℓ_2 matters much more than sparse selection

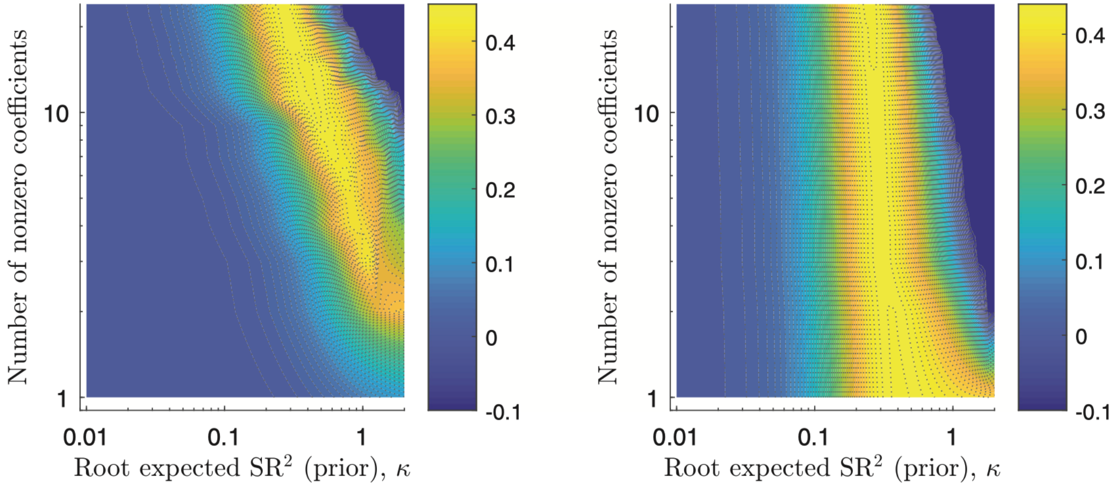
How To Pick Penalty Parameters?

- They use K-fold cross-validation
- They use $K = 3$ and split the data in a contiguous fashion
- After estimating the SDF on $K-1$ folds, they compute the out-of-sample R^2 on the holdout fold

$$R^2_{OOS} = 1 - \frac{(\bar{\mu}_K - \bar{\Sigma}_K \hat{b})^\top (\bar{\mu}_K - \bar{\Sigma}_K \hat{b})}{\bar{\mu}_K^\top \bar{\mu}_K}$$

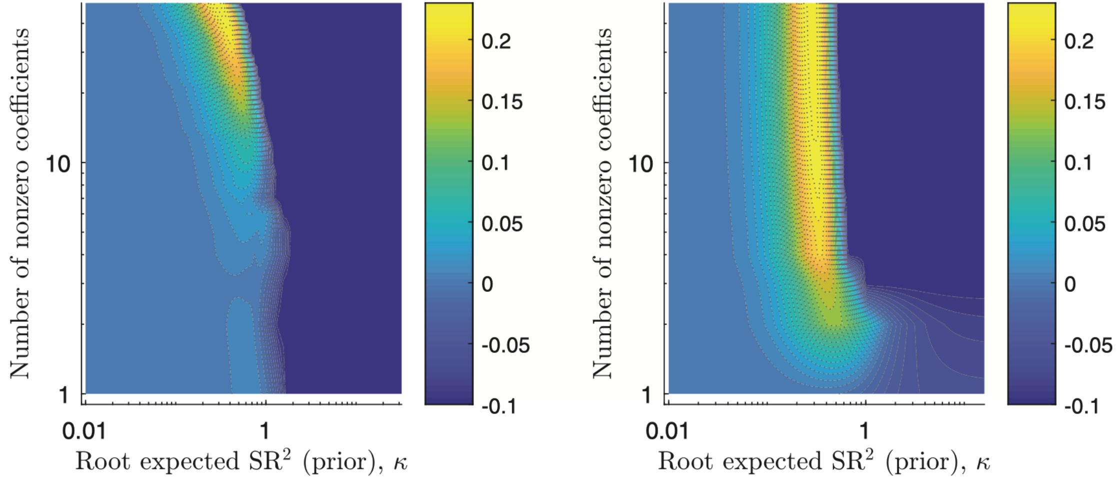
- Keep rotating and average the values

FF25 as a Sanity Check



(a) Raw Fama-French 25 portfolios **(b)** PCs of Fama-French 25 portfolios

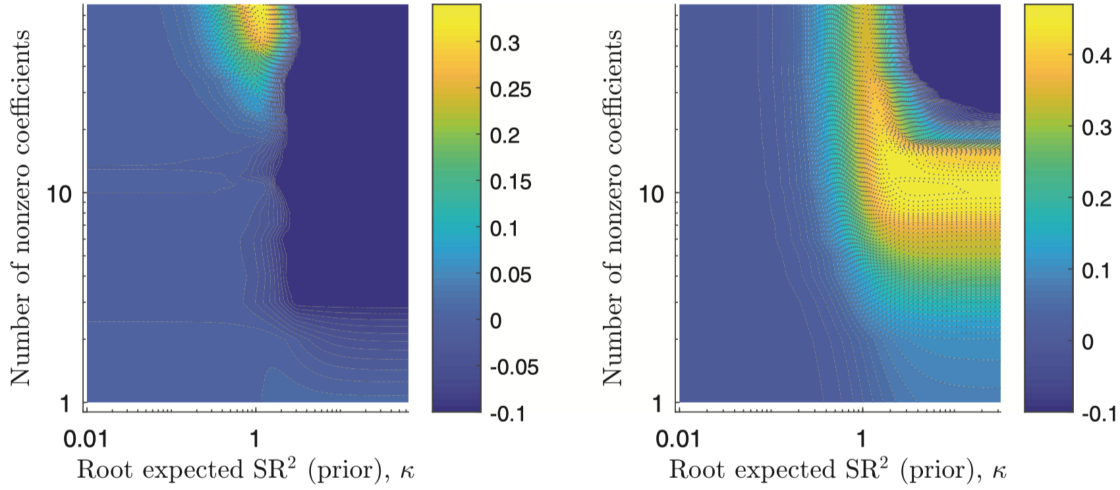
50 Anomalies From The Literature



(a) Raw 50 anomaly portfolios

(b) PCs of 50 anomaly portfolios

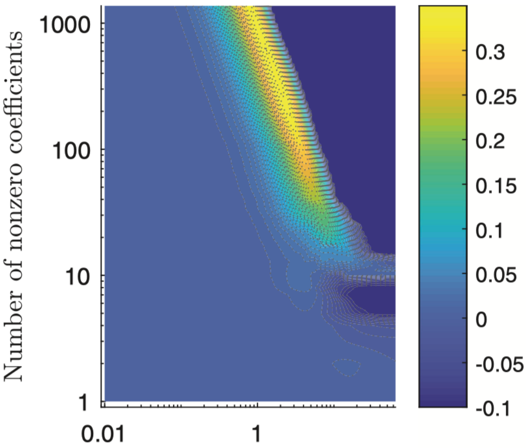
WRDS Financial Ratios: 70 Portfolios



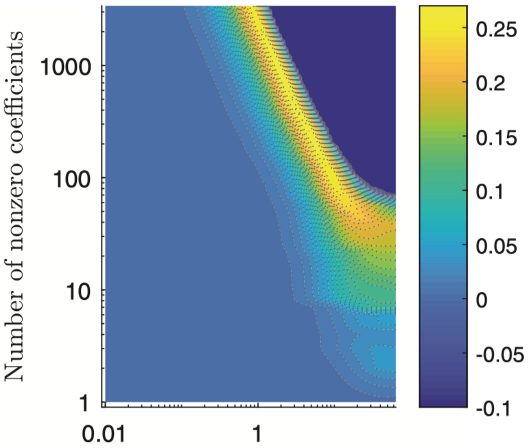
(a) Raw WFR portfolios

(b) PCs of WFR portfolios

Considering Non-Linearities: Characteristics and Their Powers



(a) 50 anomalies



(b) WFR portfolios

Contribution to the Literature

- Not another factor paper: it is a paper about how to estimate the SDF in high dimensions
- It rejects the hope that a tiny set of characteristics is enough
- It gives a formal role to shrinkage in empirical asset pricing
- One line summary: the SDF is not sparse in characteristics, but it is still shrinkable
- Their view: Asset Pricing is about many *weak* factors...

Questions?

Part II: Taming the Factor Zoo

The Question

- Suppose someone proposes a new factor today
- How should we test it against the whole existing zoo?
- A new factor should bring new pricing information, not just be correlated with the old factors
- FF3 or FF5 are convenient benchmarks, but also arbitrary
- Feng et al. (2020) propose a conservative test on purpose

Model Setup

$$m_t = \gamma_0^{-1}\left(1 - \lambda_g^\top g_t - \lambda_h^\top h_t\right)$$

$$\mathbb{E}[r_t] = \iota_n \gamma_0 + C_g \lambda_g + C_h \lambda_h$$

- γ_0: zero-beta rate (ι_n is a vector of ones)
- g_t: $d \times 1$ factors we want to test
- h_t: $p \times 1$ potentially huge set of control factors
- C_g and C_h: $n \times d$ and $n \times p$ matrices of covariances
- The goal is inference on λ_g
- The problem: h_t is large and we don't know which controls to include
- If p was small, how would you estimate λ_g?

Why Standard Two-Pass Breaks

- If p is small, ordinary two-pass methods are fine
- But in the zoo, p can be comparable to n (number of assets) or T (number of periods)
- Using all factors is infeasible or very noisy
- Using FF3 or FF5 alone creates omitted-variable bias

Sparsity as a Working Assumption

- They assume the relevant control set is approximately low-dimensional
- This is the same as saying that λ_h has many zeros
- This is the same broad "bet on sparsity" logic as Lecture 15
- This is the traditional Fama-MacBeth logic but with "endogenous" controls
- The benchmark must be selected from a large set of candidates

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- This is the traditional Fama-MacBeth logic but with "endogenous" controls
- The benchmark must be selected from a large set of candidates
- Key contribution: adapting the setup from Belloni et al. (2014) to the factor zoo
- Highly non-trivial: panel data *and* we do not directly observe covariances

Double Selection: Step 1

$$\min_{\gamma,\lambda}\left\{n^{-1}\left\|\bar{r}-\iota_n\gamma-\hat{C}_h\lambda\right\|_2^2+\tau_0 n^{-1}\|\lambda\|_1\right\}$$

- First LASSO selects controls that help explain the cross-section of average returns
- This is the "pricing ability" margin
- It looks like the natural factor-selection step
- But it is not enough for valid inference on λ_g

Double Selection: Step 2

$$\min_{\xi_j,\chi_j} \left\{ n^{-1} \left\| \hat{C}_{g,\cdot j} - \iota_n \xi_j - \hat{C}_h \chi_j^\top \right\|_2^2 + \tau_j n^{-1} \|\chi_j\|_1 \right\}$$

- Second LASSO searches for controls whose exposures line up with exposures to g_t
- This is the omitted-variable-bias margin
- It keeps factors that may look weak for pricing but are dangerous to omit
- Notice that we have d separate LASSO problems here, one for each factor we want to test

Post-Selection Estimation

$$(\hat{\gamma}_0, \hat{\lambda}_g, \hat{\lambda}_h) = \arg\min_{\gamma_0, \lambda_g, \lambda_h} \left\{ \left\| \bar{r} - \iota_n \gamma_0 - \hat{C}_g \lambda_g - \hat{C}_h \lambda_h \right\|_2^2 : \lambda_{h,j} = 0 \text{ for } j \notin \hat{I} \right\}$$

- Refit by OLS after taking the union of selected controls
- This removes the regularization bias from LASSO
- Same structure as post-double-selection in Lecture 15

Valid Inference on λ_g

Under some conditions (see the paper...):

$$\sqrt{T}\left(\hat{\lambda}_g - \lambda_g\right) \xrightarrow{d} \mathcal{N}_d(0, \Pi_1)$$

$$\Pi_1 = \lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} \sum_{s=1}^{T} \mathbb{E}\left[\left(1 - \lambda^\top v_t\right)\left(1 - \lambda^\top v_s\right) \Sigma_z^{-1} z_t z_s^\top \Sigma_z^{-1}\right], \qquad \Sigma_z \equiv \mathrm{Var}(z_t)$$

- z_t: residual from projecting g_t on the selected controls in h_t
- The asymptotic variance is long-run because dependence over time matters
- The paper also provides a Newey-West-type estimator for Π_1

Questions?

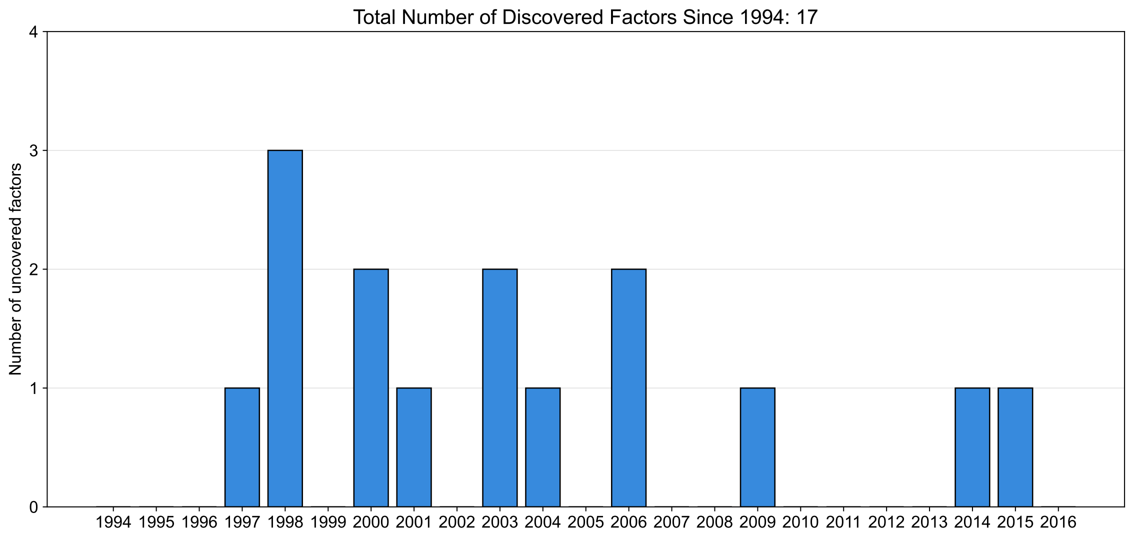
Data and Test Assets

- Data ranges from 1976 to 2017
- 150 monthly factors from 1976 to 2017
- 15 standard public factors plus 135 long-short portfolios of stocks sorted on characteristics
- Test assets: 750 characteristic-sorted test portfolios
- The large test set directly answers the LNS critique (Lewellen et al. 2010)

Factors Published in 2012-2016

Factor	λ_s (bp)	tstat (DS)
Cash holdings	-34	-0.42
HML Devil	54	1.04
Gross profitability	20	0.48
Organizational Capital	28	0.92
Betting Against Beta	35	1.45
Quality Minus Junk	73	2.03^{**}
Employee growth	43	1.36
Growth in advertising	-12	-1.18
Book Asset Liquidity	40	1.07
RMW	160	4.45^{***}
CMA	38	1.10
HXZ IA	51	2.11^{**}
HXZ ROE	77	3.37^{***}
Intermediary Risk Factor	112	2.21^{**}
Convertible debt	-15	-1.36

Recursively Testing New Factors



Contribution to the Literature

- It turns Lecture 15's double-selection logic into an asset-pricing test
- It replaces arbitrary benchmarks with selected but inference-valid benchmarks
- It responds directly to LNS and complements HLZ (Lewellen et al. 2010; Harvey et al. 2016)
- It makes factor discovery conservative on purpose

Questions?

Taking Stock

- Lectures 12 and 13 built factor models
- Lecture 14 showed the zoo and the weakness of old testing standards
- Lecture 15 gave us shrinkage and valid post-selection inference
- Lecture 16: modern empirical asset pricing means regularization plus tough benchmarks
- A factor is not interesting because it has a story; it must survive high-dimensional comparison

Readings and References i

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